

Suburban PV

Solar PV is becoming an increasingly popular option for generating electricity in urban and suburban homes. Three years ago Niels Wolter of Madison, Wisconsin, installed a 1.3 kW PV system on his family's three-bedroom masonry home. The system provides about 80% of their electricity needs, which average out to around 2,000 kWh per year.

Wolter chose a pole-mounted, dual-axis tracking PV system, which is grid connected, for installation in their sunny, suburban backyard rather than on the roof of their home. A tracking mechanism on the pole keeps the Kyocera-manufactured PV panels pointed at the sun to maximize electricity production. While it's difficult to avoid some shading from neighborhood trees in an urban setting, the PV system is only about 12% shaded.

The PV system is interconnected to the utility grid so it doesn't need to meet all the family's electricity needs. "However," says Wolter, "if you install a system that's slightly undersized, it can inspire you to become even more energy efficient."

Purchasing a PV system was not Wolter's first step in taking control of his home's energy use. Beginning in 2000, Wolter installed CFLs throughout the house. In 2002, he purchased an energy-efficient freezer, and in 2003, an Energy Star refrigerator. The savings in the electricity bill alone let the freezer pay for itself in less than four years. After conducting a blower door test, Wolter air sealed and installed five new double-pane windows, which are used with storm windows and insulated window shades. Wolter also uses an Energy Star laptop computer, washes clothes in cold water, line dries laundry, and keeps the indoor temperature low in winter, while using a programmable thermostat, attic fan, and natural gas water heating. Wolter and his son have also developed some simple energy-saving habits around the house, such as turning off lights and appliances when they're not needed.

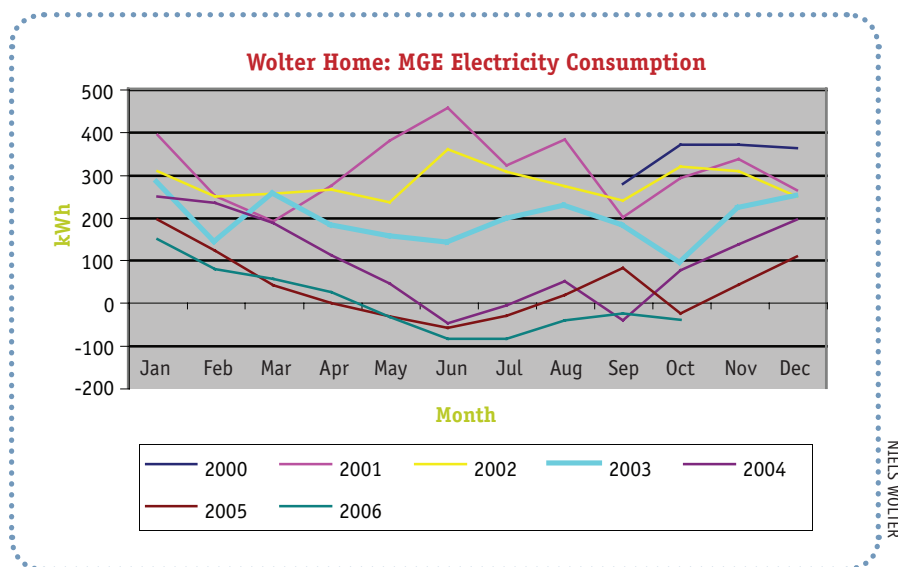


Figure 1. The Wolter's PV system went online in June 2004, and since then they have cut electricity use by almost 50%.

For Wolter, installing a PV system was the logical next step in honoring his commitment to a clean-energy future; the system went online in June 2004. Wolter and his son had never been heavy consumers of electricity; now they have cut their use by close to 50% (see Figure 1). With the new PV panels, which should last 25–50 years, the Wolters's electricity bill will be less than 5% of what the average Wisconsin family spends. This cut in electricity use meant that the PV system could be downsized. The Wolters invested less than \$1,000 in energy efficiency and were able to reduce the cost of their PV system by about \$7,500.

After incentives, the installed PV system cost about \$7,600, with a simple payback of 39 years at today's utility rates. Wolter says that lining up the money was the greatest challenge; much of the actual installation was easy. "We have great installers, and the utility, Madison Gas and Electric, is easy to work with, so there were no problems."

The projected annual electricity usage and cost after installation is 250 kWh/per year, which at current rates totals \$130 (75% of the annual cost consists of fixed charges for connection). Furthermore, by using renewable energy, the Wolters drastically cut the amount of pollution and greenhouse gases their energy use produces. At its current generation output, the system

will offset the equivalent of 1,550 lb of coal burned per year, with an avoided pollution per year of 3,479 lb of CO₂; 14 lb of sulfur dioxide (SO₂); and 7.3 lb nitrogen oxide (NO_x). These calculations are based on assumed rates of 1.988 lb CO₂/kWh; 7.99 lb SO₂/MWh; and 4.16 lb NO.

In 2005, the Wolters's house was on a solar-home tour, during which it was seen by over 100 people, including a few neighbors. Since then, says Wolter, "one neighbor now has solar electric on his house too, and my mailman will soon be installing a solar water-heating system. All the neighbors seem to like it."

In June 2006, Wolter installed a solar hot water system. When asked if there were any regional challenges to installing solar water and PV systems in Wisconsin, Wolter says that homeowners don't know too much about solar options in residential homes. "At the moment more people need to be aware that solar works and is available."

"Anyone can take these steps," says Wolter. "You don't need to get a giant system. Besides, it's a lot of fun, and every solar-electric system makes Wisconsin's future a little greener!"

—Elka Karl

Elka Karl is an associate editor of *Home Energy*. This article was adapted from a Wisconsin Focus on Energy case study.